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Chiang et al.

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(54) **FOLDABLE ELECTRONIC DEVICE**

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G06F 1/16 (2006.01)
F16C 11/04 (2006.01)

(52) **U.S. Cl.**
CPC ***F16C 11/04*** (2013.01); ***G06F 1/1616*** (2013.01); ***G06F 1/1681*** (2013.01)

(58) **Field of Classification Search**
None
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

11,359,425	B2 *	6/2022	Chang	E05D 3/18
11,775,029	B2 *	10/2023	Yuan	G06F 1/181 361/679.27
11,782,487	B2 *	10/2023	Shen	G06F 1/1681 361/679.27
11,797,061	B2 *	10/2023	Hazama	G06F 1/1681
11,924,987	B2 *	3/2024	Lee	H04M 1/022
11,927,991	B2 *	3/2024	Torres	E05D 3/16
11,983,045	B2 *	5/2024	Lin	G06F 1/1681
12,108,551	B2 *	10/2024	Jung	H05K 5/0226
12,117,877	B2 *	10/2024	Zhang	H04M 1/0216
12,124,293	B2 *	10/2024	Chen	G06F 1/1675
12,204,380	B2 *	1/2025	Park	G06F 1/1618
2018/0073613	A1 *	3/2018	Chen	F16H 25/18
2018/0119467	A1 *	5/2018	Liu	E05D 11/06
2022/0147112	A1 *	5/2022	Yu	F16C 11/04

FOREIGN PATENT DOCUMENTS

CN	210830137	6/2020
CN	113124042	7/2021
TW	I694214	5/2020

* cited by examiner

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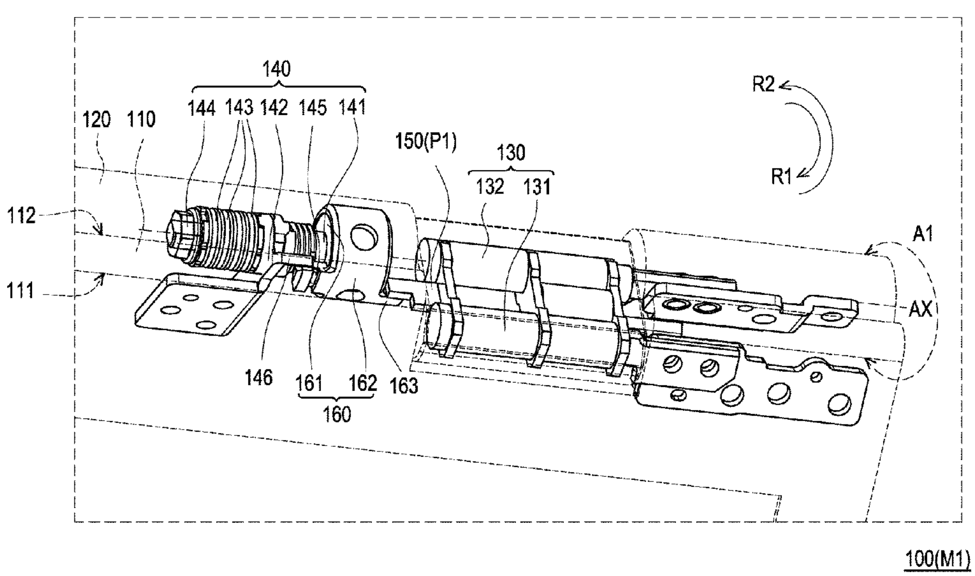
(57) **ABSTRACT**

A foldable electronic device includes a first body, a second body, a first hinge module, a second hinge module, a driving sheet, and a one-way bearing. The first body has a display surface and a back surface. The first hinge module has a first shaft pivotally connected to the first body and a second shaft pivotally connected to the second body. The second shaft has a virtual axis. The second hinge module is disposed to the second body and has a rotating shaft. The rotating shaft is disposed corresponding to the virtual axis. The driving sheet is located between the first shaft and the second shaft, and is located between the first hinge module and the second hinge module. The one-way bearing is rotatably disposed around the rotating shaft. The one-way bearing has a bearing stop portion corresponding to the driving sheet.

8 Claims, 8 Drawing Sheets



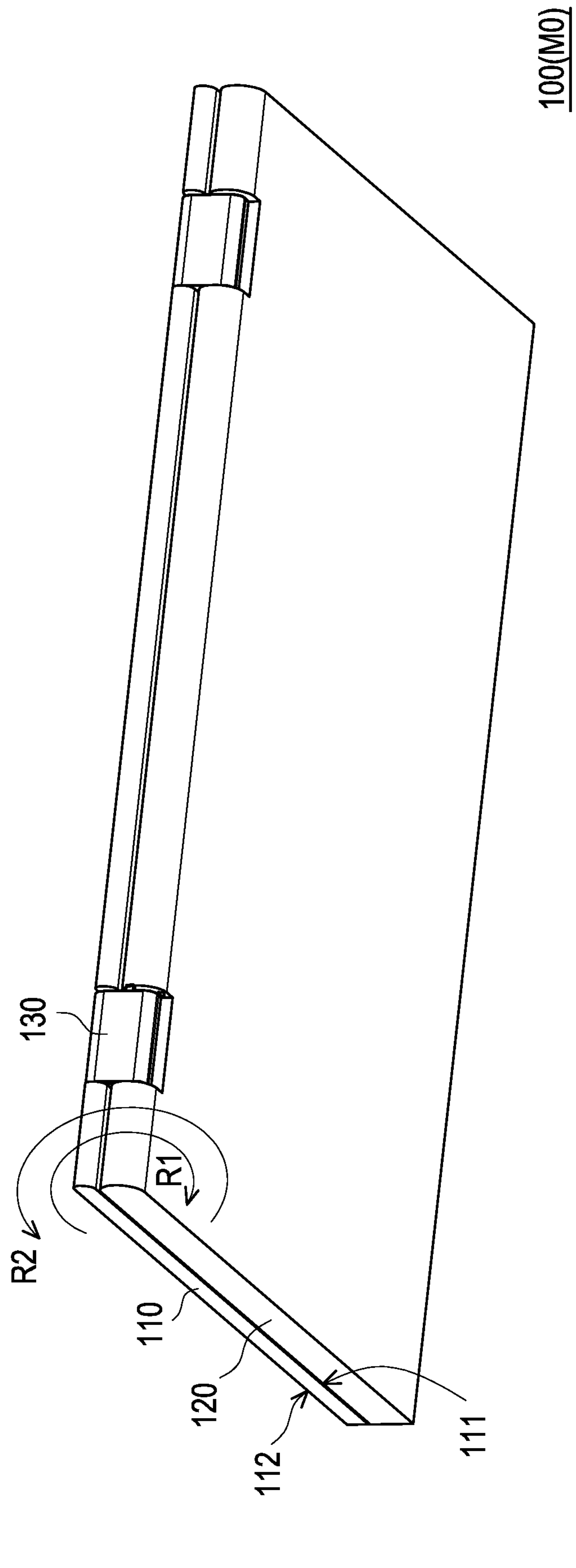


FIG. 1A

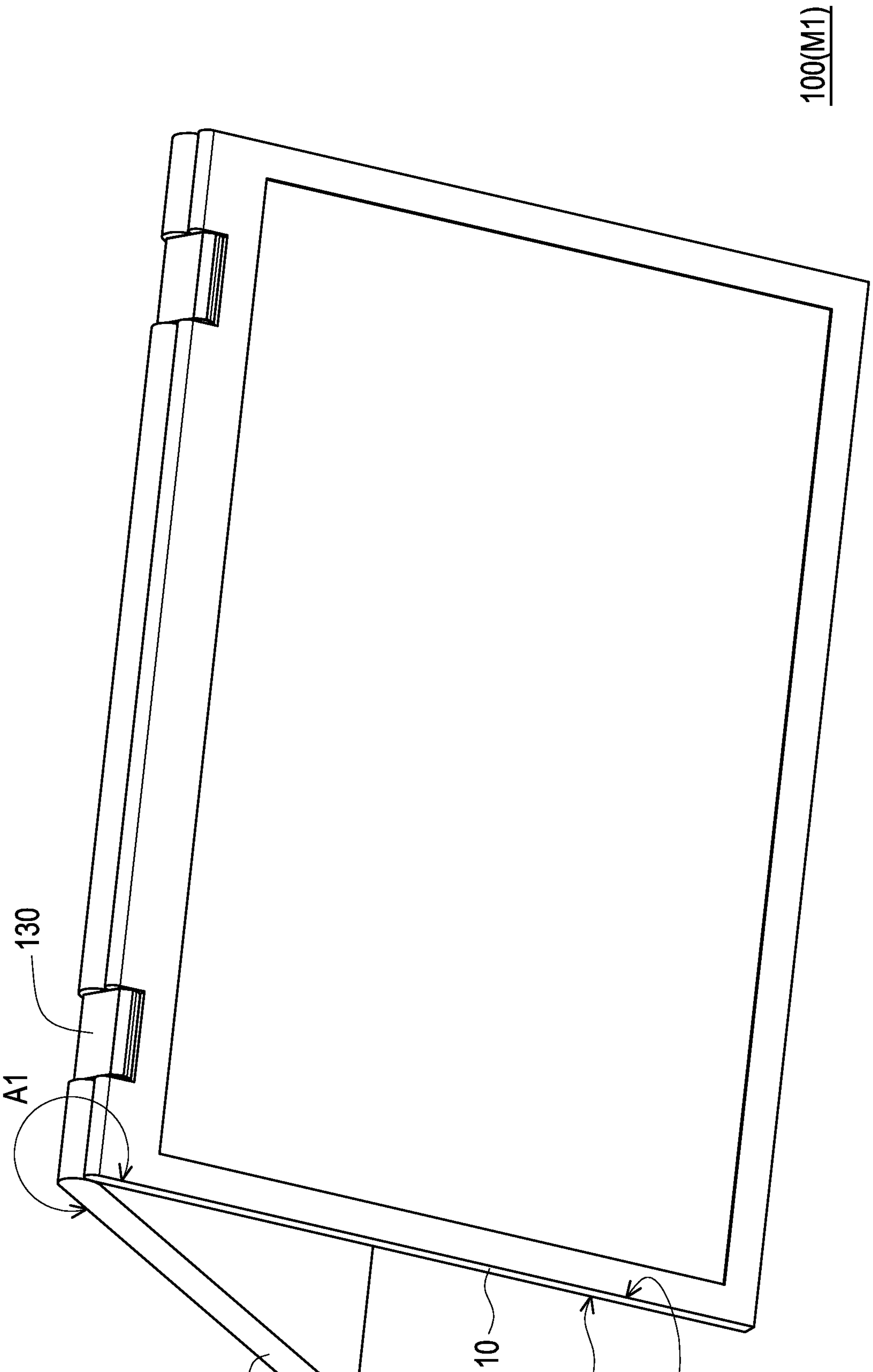


FIG. 1B

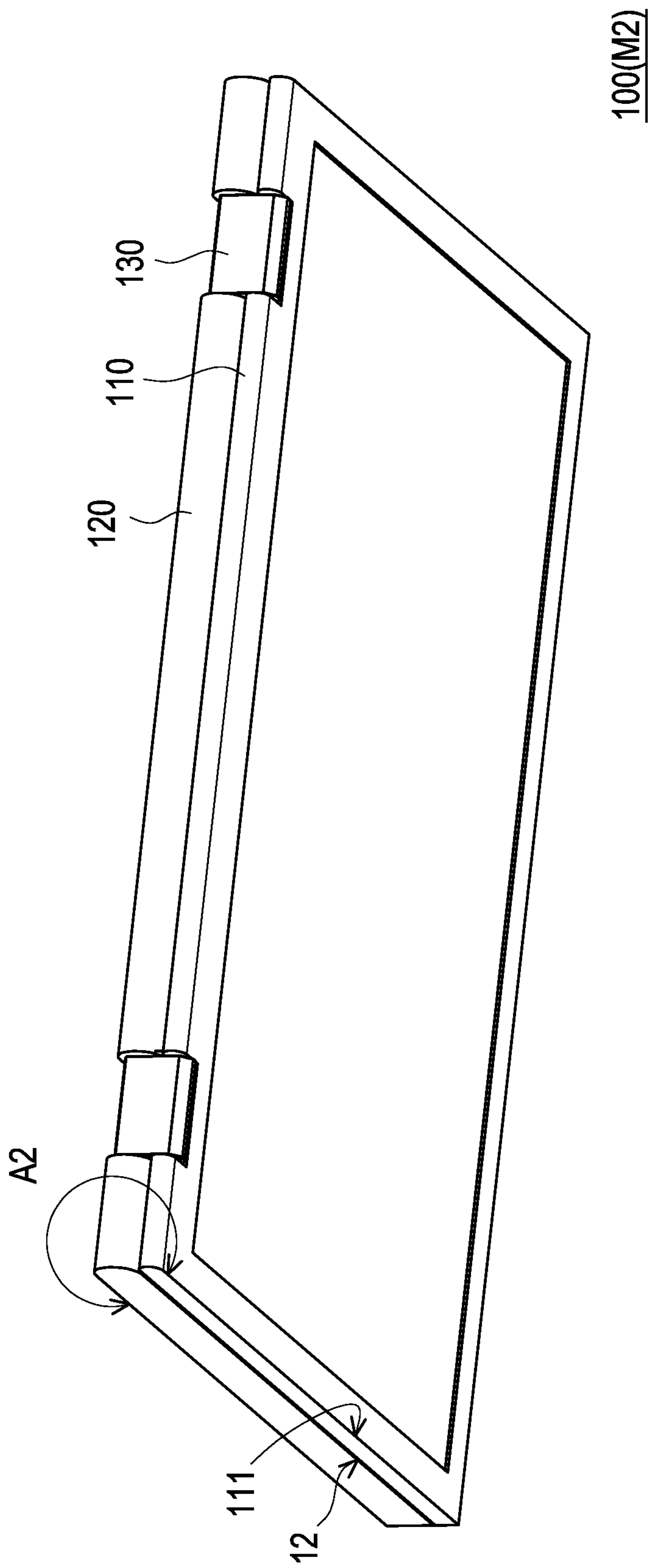
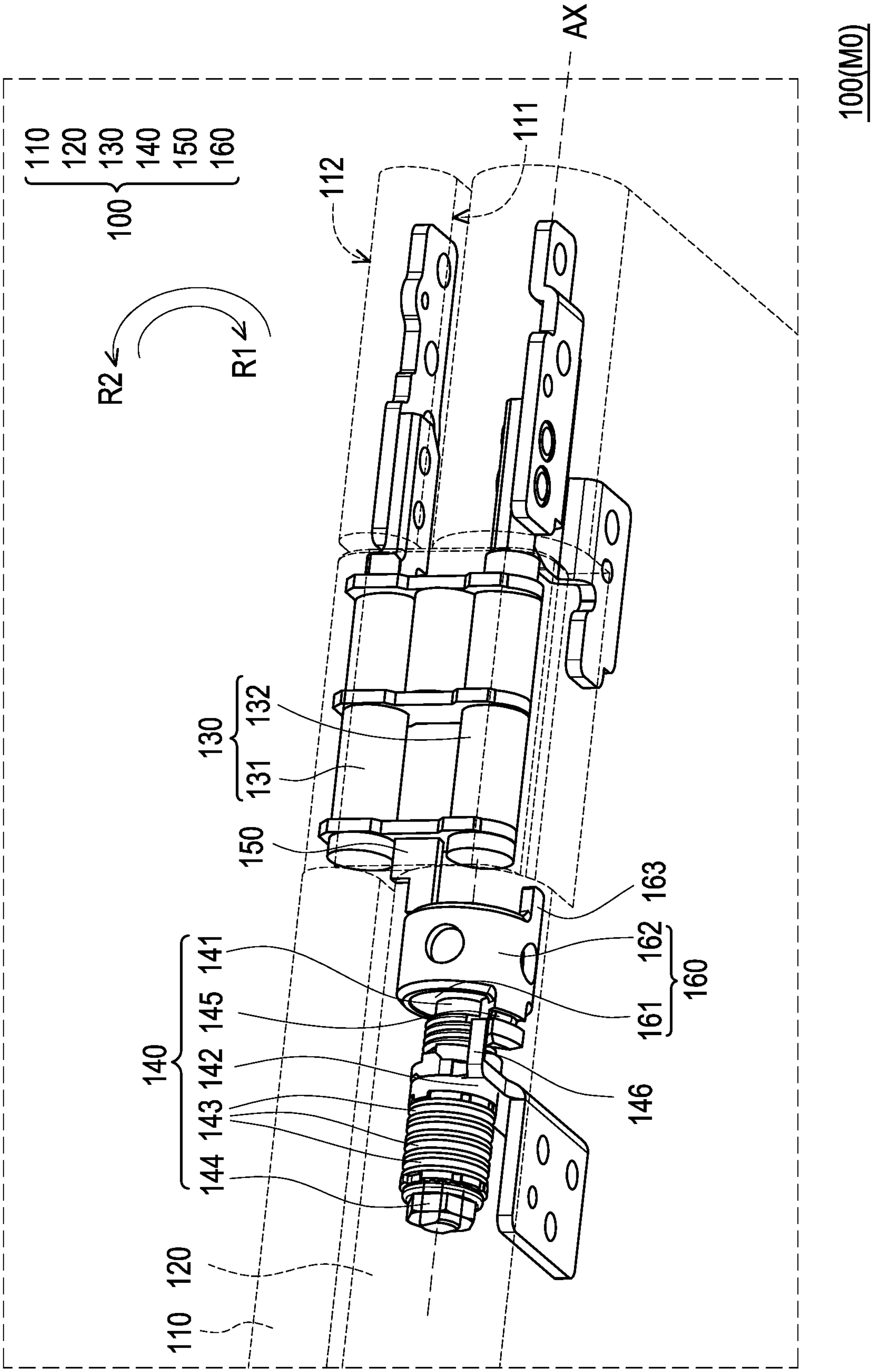
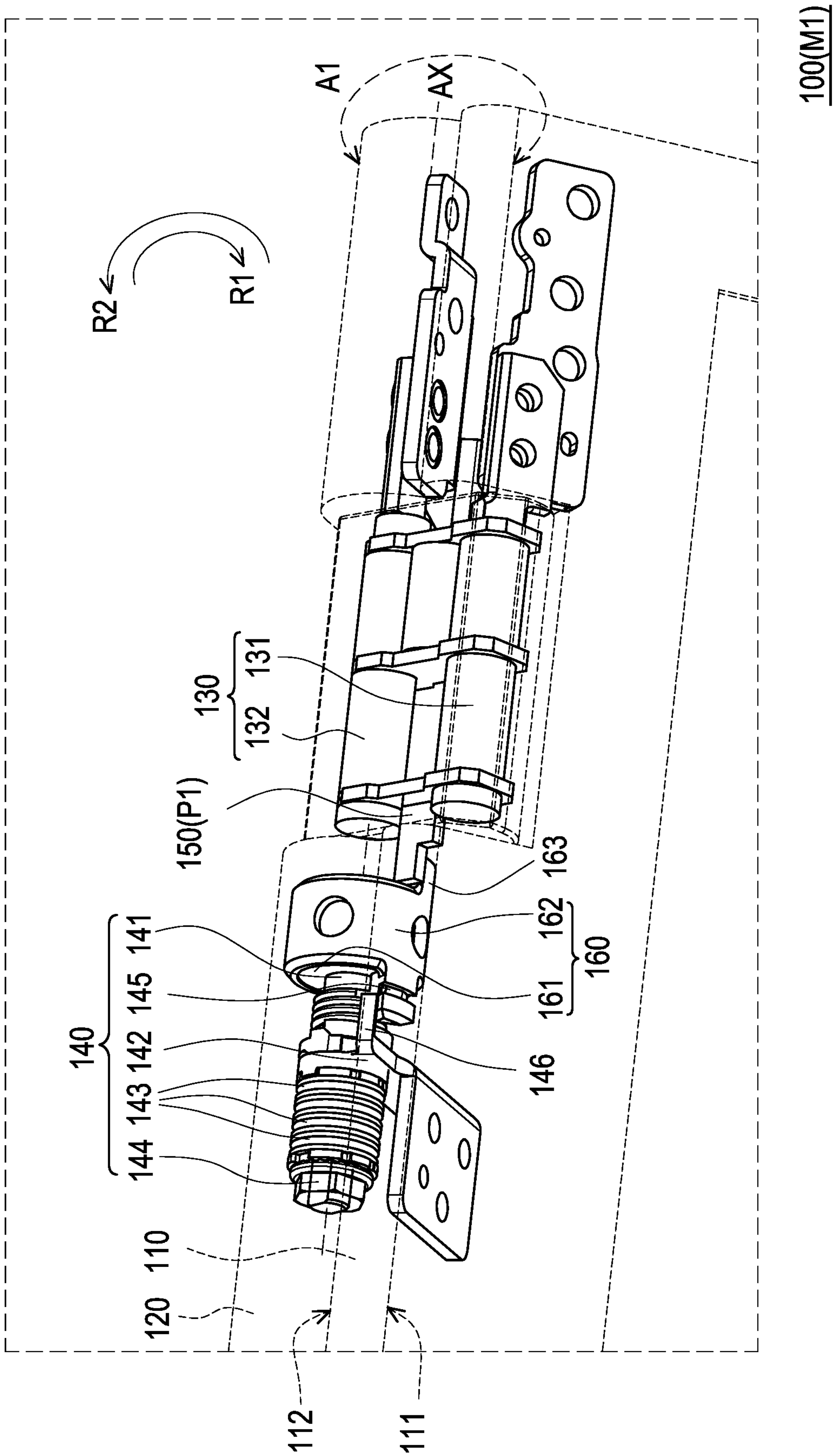


FIG. 1C



AX
100(M0)
100
110
120
130
140
150
160
111
112
R1
R2
130
131
132
150
163
162
160
161
140
141
145
142
143
144
146
120
110

FIG. 2A



100(M1)
A1
AX
R2
R1
130
131
132
150(P1)
163
162
160
161
140
141
145
142
146
143
144
110
120
112
111

FIG. 2B

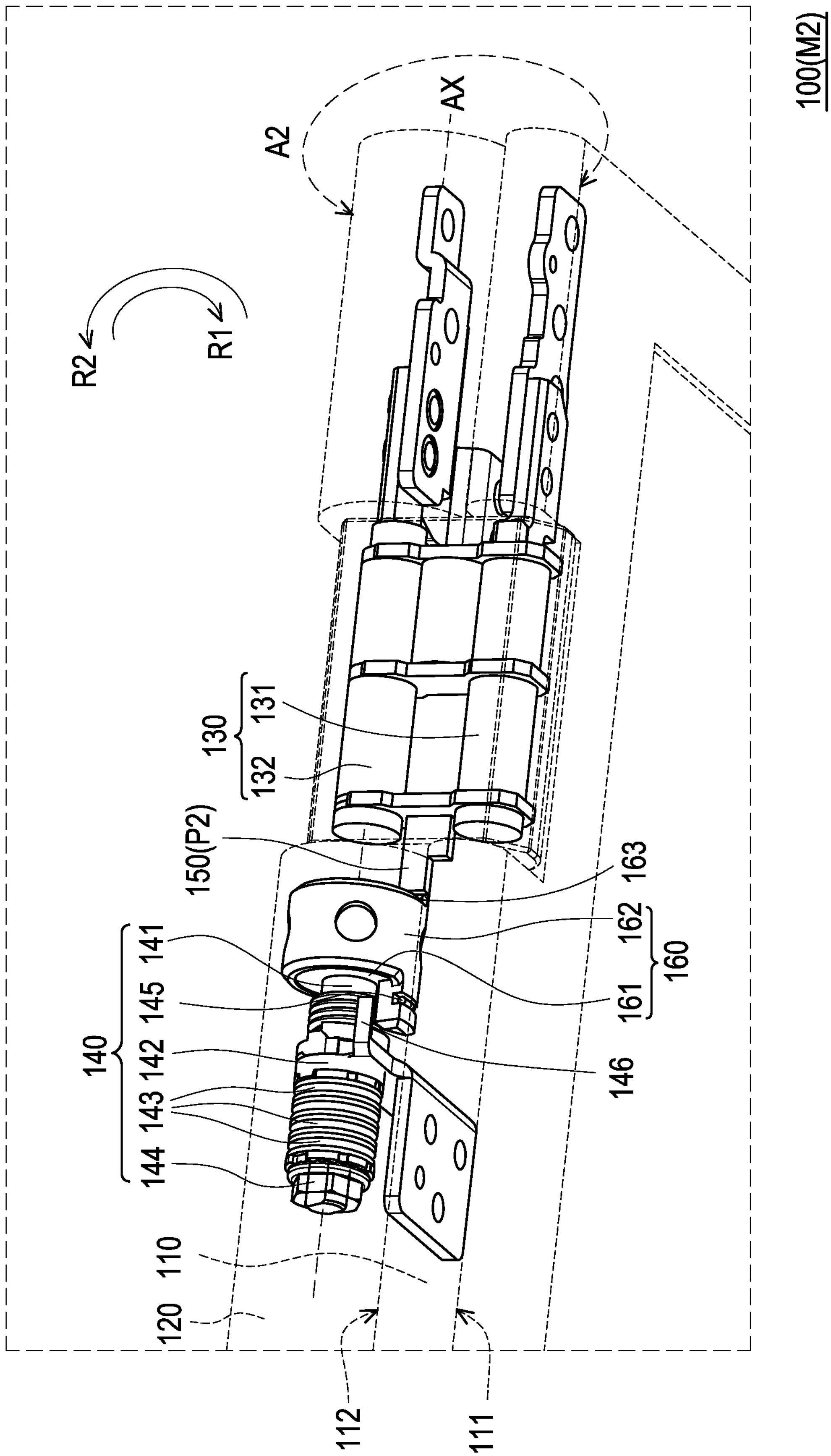

100(M2)
AX
A2
R2
R1
130
131
132
150(P2)
163
162
161
160
140
141
145
142
143
144
146
110
120
112
111

FIG. 2C

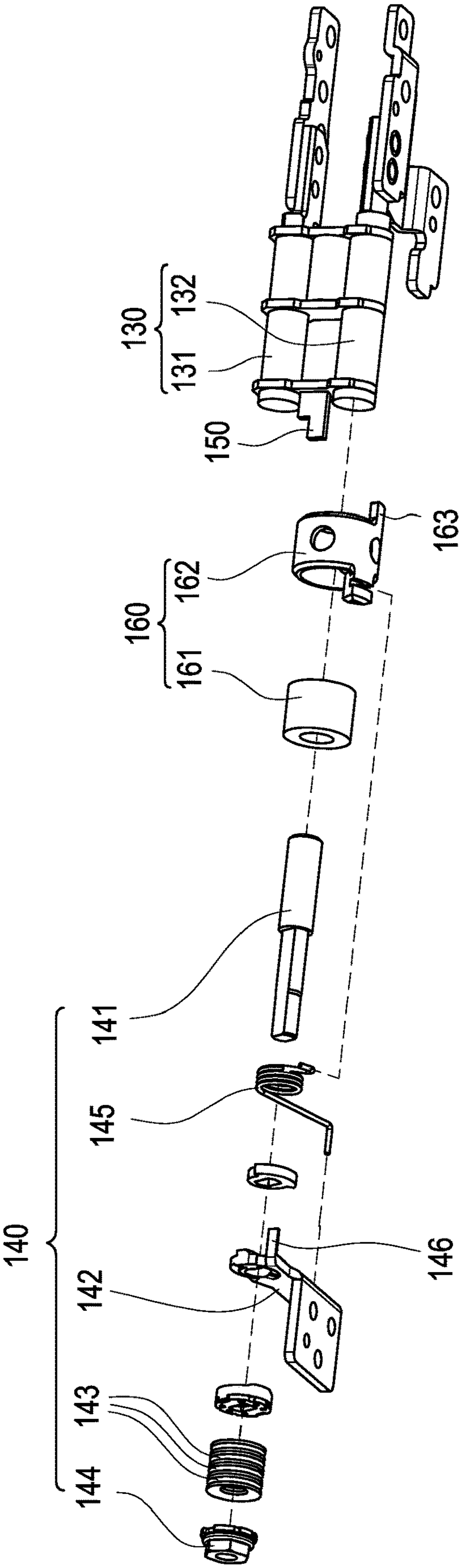


FIG. 3

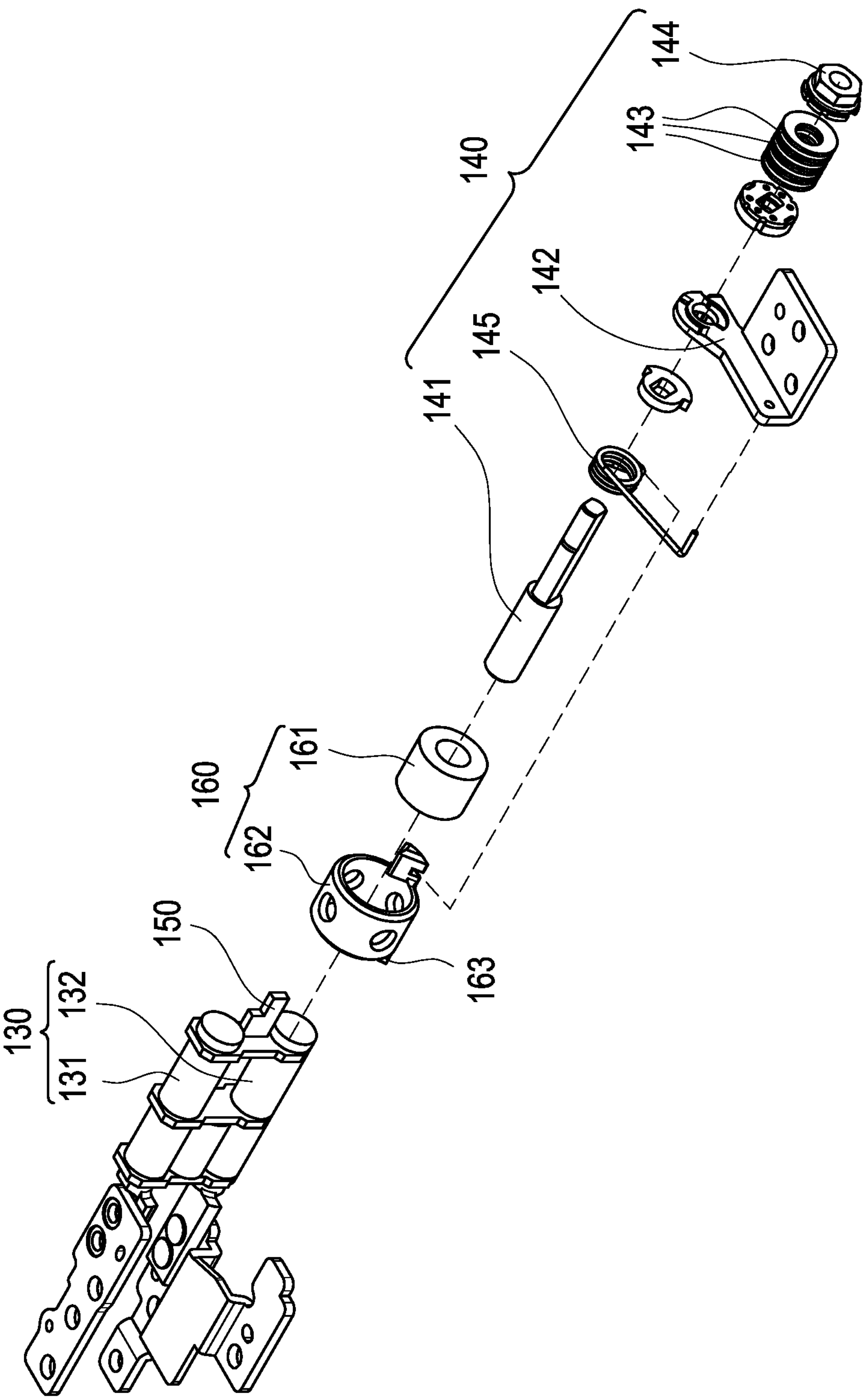


FIG. 4

FOLDABLE ELECTRONIC DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

This application claims the priority benefit of Taiwan application serial no. 111211031, filed on Oct. 7, 2022. The entirety of the above-mentioned patent application is hereby incorporated by reference herein and made a part of this specification.

BACKGROUND

Technical Field

The disclosure relates to an electronic device, and in particular relates to a foldable electronic device.

Description of Related Art

Generally speaking, in a 360-degree foldable electronic device, when the display body and the system body are unfolded and folded with each other, there may be various operation modes, such as a notebook computer mode, a tablet mode, a tent mode, and a drawing mode. Moreover, the torque provided by the hinge module pivotally connected between the display body and the system body is mostly constant, so that the display body and the system body may rotate smoothly during the unfolding and folding process.

However, in the drawing mode, the user often applies extra force to the display body to draw and write via touch control, such that the torque value originally set by the hinge module cannot support the force of drawing and writing via touch control. Alternatively, the torque of the hinge module needs to be increased, thereby making it difficult for the user to flip the electronic device in other modes.

SUMMARY

In this disclosure, a foldable electronic device, which includes a first body, a second body, a first hinge module, a second hinge module, a driving sheet, and a one-way bearing are provided. The first body has a display surface and a back surface. The first hinge module has a first shaft and a second shaft. The first shaft is pivotally connected to the first body. The second shaft is pivotally connected to the second body. The second shaft has a virtual axis. The second hinge module is disposed on the second body and has a rotating shaft. The rotating shaft is disposed corresponding to the virtual axis. The driving sheet is located between the first shaft and the second shaft. The driving sheet is located between the first hinge module and the second hinge module. The one-way bearing is rotatably disposed around the rotating shaft. The one-way bearing has a bearing stop portion corresponding to the driving sheet. When the first body is flipped relative to the second body, the driving sheet rotates to a first position along a first rotation direction and contacts the bearing stop portion, and drives the one-way bearing and the rotating shaft to rotate relative to the second body simultaneously, so that the back surface faces the second body.

Based on the above, in the foldable electronic device of this disclosure, the first body is rotated 360 degrees relative to the second body through the first hinge module, and through the structural design of the second hinge module, the driving sheet, and the one-way bearing, such that the second hinge module in the drawing mode provides additional torque only after the first body rotates relative to the second body along the first rotation direction and unfolds to a first angle (i.e., the driving sheet rotates to the first position along the first rotation direction to contact the bearing stop portion). Therefore, in the drawing mode, the foldable electronic device supports the force of the hand drawing and writing via touch control applied along the first rotation direction, and prevents the user from feeling difficulty in flipping the electronic device in other modes caused by the increase of the torque value.

BRIEF DESCRIPTION OF THE DRAWING

FIG. **1**A to FIG. **1**C are three-dimensional views of a foldable electronic device in a folding mode, a drawing mode, and a tablet mode of an embodiment of the disclosure.

FIG. **2**A is a partially enlarged perspective view of FIG. **1**A.

FIG. **2**B is a partially enlarged perspective view of FIG. **1**B.

FIG. **2**C is a partially enlarged perspective view of FIG. **1**C.

FIG. **3** is a partial component exploded view of the first hinge module, the second hinge module, the driving sheet, and the one-way bearing of FIG. **2**A.

FIG. **4** is a partial component exploded view of FIG. **3** from another perspective.

DETAILED DESCRIPTION OF DISCLOSED EMBODIMENTS

FIG. **1**A to FIG. **1**C are three-dimensional views of a foldable electronic device **100** in a folding mode M**0**, a drawing mode M**1**, and a tablet mode M**2** of an embodiment of the disclosure.

Referring to FIG. **1**A to FIG. **1**C, the foldable electronic device **100** is, for example, a notebook computer. The foldable electronic device **100** is in the folding mode M**0** (FIG. **1**A) to facilitate carrying and storage, and the foldable electronic device **100** may also be in the unfolded state. In the unfolded state, the foldable electronic device is unfolded by a first angle A**1** (e.g., 315 degrees) to be the drawing mode M**1** (FIG. **1**B), or unfolded to a second angle A**2** (e.g., 360 degrees) to be the tablet mode M**2** (FIG. **1**C) according to the usage requirements.

The torque value of the foldable electronic device **100** unfolded from the first angle A**1** to the second angle A**2** is greater than the torque value of the foldable electronic device **100** unfolding from 0 degrees to the first angle A**1** and the torque value of the foldable electronic device **100** closing from the second angle A**2** to 0 degrees.

Referring to FIG. **2**A to FIG. **2**C, the foldable electronic device **100** of the disclosure includes a first body **110**, a second body **120**, and a first hinge module **130**. The first hinge module **130** is pivotally connected to the first body **110** and the second body **120**, so that the first body **110** rotates 360 degrees relative to the second body **120** through the first hinge module **130**.

In one embodiment, the first body **110** is, for example, a display body, and the second body **120** is, for example, a system body, but not limited thereto.

In one embodiment, the first hinge module **130** is, for example, a gear hinge, so that the first body **110** and the second body **120** rotates synchronously with respect to each other, but not limited thereto. In other embodiments, the first hinge module **130** is a switch hinge, which achieves multi-stage rotation by means of alternating rotation, but is not limited thereto.

In one embodiment, the first body **110** has a display surface **111** and a back surface **112** opposite to each other. In the folding mode M**0** (FIG. **1**A), the display surface **111** is located between the second body **120** and the back surface **112**, while in the drawing mode M**1** (FIG. **1**B) or the tablet mode M**2** (FIG. **1**C), the back surface **112** is located between the second body **120** and the display surface **111**, but not limited thereto.

In one embodiment, as shown in FIG. **1**A, the rotation direction of the first body **110** unfolding relative to the second body **120** is the first rotation direction R**1**, and the rotation direction of the first body **110** closing relative to the second body **120** is the second rotation direction R**2**, but not limited thereto.

Referring to FIG. **2**A to FIG. **2**C, the first hinge module **130** has a first shaft **131** and a second shaft **132**. The first shaft **131** is pivotally connected to the first body **110**. The second shaft **132** is pivotally connected to the second body **120**, and the second shaft **132** has a virtual axis AX.

Referring to FIG. **2**A to FIG. **2**C, the foldable electronic device **100** of the disclosure further includes a second hinge module **140**, a driving sheet **150**, and a one-way bearing **160**. The second hinge module **140** is disposed on the second body **120** and has a rotating shaft **141**, and the rotating shaft **141** is disposed corresponding to the virtual axis AX. The driving sheet **150** is located between the first shaft **131** and the second shaft **132**, is located between the first hinge module **130** and the second hinge module **140**, and rotates together with the first body **110** around the virtual axis AX. The one-way bearing **160** is rotatably disposed around the rotating shaft **141** and has a bearing stop portion **163** corresponding to the driving sheet **150**.

Referring to FIG. **2**A to FIG. **4**, the second hinge module **140** includes a fixing member **142**, multiple torque members **143**, a locking attachment **144**, and an elastic member **145**. The fixing member **142** is connected to the second body **120**, and disposed around the rotating shaft **141**. The torque members **143** are disposed around the rotating shaft **141**. The locking attachment **144** is locked to the rotating shaft **141** to sandwich the torque members **143** between the fixing member **142** and the locking attachment **144**. In this way, when the rotating shaft **141** rotates relative to the fixing member **142** disposed on the second body **120**, the second hinge module **140** generates torque.

In one embodiment, the torque value provided by the second hinge module **140** is, for example, 0.7 times the torque value provided by the first hinge module **130**, but not limited thereto.

Referring to FIG. **3** and FIG. **4**, the elastic member **145** is connected between the fixing member **142** and the one-way bearing **160**, and the fixing member **142** has a hinge stop portion **146** (FIG. **3**) corresponding to the one-way bearing **160**, so that the elastic member **145** applies returning force to the one-way bearing **160**, and when the one-way bearing **160** is rotated in the second rotation direction R**2** relative to the rotating shaft **141**, the one-way bearing **160** may come into contact with the hinge stop portion **146** and stop the rotation (FIG. **2**B).

In this way, through the structural design of the elastic member **145** and the hinge stop portion **146**, the one-way bearing **160** rotates to the position where it contacts the hinge stop portion **146** (FIG. **2**B) along the second rotation direction R**2** is the same as the position where the driving sheet **150** rotates to the first position P**1** along the first rotation direction R**1** and contacts the bearing stop portion **163** (FIG. **2**B), so that the one-way bearing **160** is smoothly returned to the original position (i.e., the position in contact with the driving sheet **150** at the first position P**1**).

Referring to FIG. **2**A to FIG. **4**, the one-way bearing **160** includes an inner ring member **161** and an outer ring member **162**. The inner ring member **161** is disposed around the rotating shaft **141**, the outer ring member **162** is disposed around the inner ring member **161**, and the outer ring member **162** has bearing stop portion **163**.

The outer ring member **162** is combined with the inner ring member **161** by means of, for example, interference-fitting, adhesive bonding, screw locking, etc., so that the outer ring member **162** is tightly fitted to the inner ring member **161** in the first rotation direction R**1**, and the outer ring member **162** rotates relative to the inner ring member **161** in the second rotation direction R**2**, but not limited thereto.

In this way, through the structural design of the driving sheet **150** and the one-way bearing **160**, the driving sheet **150** may only drive the one-way bearing **160** and the rotating shaft **141** together to rotate along the first rotation direction R**1** under the contact with the bearing stop portion **163** (i.e., the driving sheet **150** rotates from the first position P**1** in FIG. **2**B to the second position P**2** in FIG. **2**C), such that the second hinge module **140** may only provide another torque when the first body **110** relative to the second body **120** is in a specific rotation direction and a specific unfolding angle range.

That is, as shown in FIG. **1**A and FIG. **2**A, before the first body **110** is unfolded to the first angle A**1** (FIG. **2**B) relative to the second body **120** along the first rotation direction R**1**, the driving sheet **150** has not yet rotated to the first position P**1** (FIG. **2**B) and is not in contact with the bearing stop portion **163**, therefore the second hinge module **140** cannot be driven to generate torque. In other words, during the transition from the folding mode M**0** to the drawing mode M**1**, the torque of the foldable electronic device **100** is only provided by the first hinge module **130**.

Next, as shown in FIG. **1**B and FIG. **2**B, when the first body **110** is unfolded to the first angle A**1** relative to the second body **120** along the first rotation direction R**1** and is in the drawing mode M**1**, the driving sheet **150** is simultaneously driven to rotate to the first position P**1** to contact the bearing stop portion **163**.

Afterwards, as shown in FIG. **1**B, FIG. **1**C, FIG. **2**B, and FIG. **2**C, if the user applies a force to the display surface **111** of the first body **110**, the first body **110** is driven to rotate along the first rotation direction R**1**, such that the driving sheet **150** simultaneously drives the one-way bearing **160** and the rotating shaft **141** together to rotate relative to the second body **120** along the first rotation direction R**1** to generate a torque, so that the back surface **112** faces the second body **120**. Therefore, during the transition from the drawing mode M**1** to the tablet mode M**2**, the torque of the foldable electronic device **100** is jointly provided by the first hinge module **130** and the second hinge module **140**.

Finally, as shown in FIG. **1**A to FIG. **2**C, when the operation mode of the foldable electronic device **100** is to be adjusted, the first body **110** is closed from the second angle A**2** (FIG. **2**C) relative to the second body **120** along the second rotation direction R**2**, such that the driving sheet **150** is driven together to rotate along the second rotation direction R**2**.

At this time, the elastic member **145** applies the returning force to the one-way bearing **160** simultaneously, so that the outer ring member **162** of the one-way bearing **160** rotates from the second position P**2** (FIG. **2**C) relative to the rotating shaft **141** in the second rotation direction R**2**, such that the rotating shaft **141** cannot be driven to rotate together to generate torque. Until the driving sheet **150** rotates and returns to the first position P**1** (FIG. **2**B), and the one-way bearing **160** contacts the hinge stop portion **146** (FIG. **2**B) and stops rotating simultaneously, the driving sheet **150** is separated from the bearing stop portion **163** (FIG. **2**A).

Therefore, during the transition from the tablet mode M**2**, through the drawing mode M**1**, to the folding mode M**0**, the torque of the foldable electronic device **100** is only provided by the first hinge module **130**.

In this way, through the structural design of the second hinge module **140**, the driving sheet **150**, and the one-way bearing **160**, the second hinge module **140** may only provide another torque after the first body **110** rotates relative to the second body **120** along the first rotation direction R**1** and is unfolded to the first angle A**1** (between the drawing mode M**1** and the tablet mode M**2**). Therefore, the foldable electronic device supports the force of the hand drawing and writing via touch control (i.e., the external force applied to the display surface **111** of the first body **110** along the first rotation direction R**1**), and prevents the user from having a difficult time in flipping the electronic device in other modes caused by the increase of the torque value.

To sum up, in the foldable electronic device of this disclosure, the first body is rotated 360 degrees relative to the second body through the first hinge module, and through the structural design of the second hinge module, the driving sheet, and the one-way bearing, such that the second hinge module in the drawing mode provides additional torque only after the first body rotates relative to the second body along the first rotation direction and unfolds to a first angle (i.e., the driving sheet rotates to the first position along the first rotation direction to contact the bearing stop portion). Therefore, in the drawing mode, the foldable electronic device supports the force of the hand drawing and writing via touch control applied along the first rotation direction, and prevents the user from feeling difficulty in flipping the electronic device in other modes caused by the increase of the torque value.

Although the disclosure has been described in detail with reference to the above embodiments, they are not intended to limit the disclosure. Those skilled in the art should understand that it is possible to make changes and modifications without departing from the spirit and scope of the disclosure. Therefore, the protection scope of the disclosure shall be defined by the following claims.

What is claimed is:

1. A foldable electronic device, comprising:

a first body, having a display surface and a back surface;

a second body;

a first hinge module, having a first shaft and a second shaft, wherein the first shaft is pivotally connected to the first body, the second shaft is pivotally connected to the second body, and the second shaft has a virtual axis;

a second hinge module, disposed on the second body and having a rotating shaft, wherein the rotating shaft is disposed corresponding to the virtual axis;

a driving sheet, located between the first shaft and the second shaft, wherein the driving sheet is located between the first hinge module and the second hinge module; and

a one-way bearing, rotatably disposed around the rotating shaft and having a bearing stop portion corresponding to the driving sheet;

wherein when the first body is flipped relative to the second body, the driving sheet rotates to a first position along a first rotation direction and contacts the bearing stop portion, and the one-way bearing and the rotating shaft are driven to rotate simultaneously relative to the second body, so that the back surface faces the second body,

wherein the second hinge module further comprises:

a fixing member, connected to the second body and disposed around the rotating shaft;

a plurality of torque members, disposed around the rotating shaft; and

a locking attachment, locked to the rotating shaft to sandwich the torque members between the fixing member and the locking attachment.

2. The foldable electronic device according to claim **1**, wherein when the one-way bearing and the rotating shaft rotate simultaneously relative to the second body along the first rotation direction, a torque is generated, and the one-way bearing rotates relative to the rotating shaft along a second rotating direction opposite to the first rotating direction until the driving sheet returns to the first position.

3. The foldable electronic device according to claim **1**, wherein the second hinge module further comprises an elastic member connected between the fixing member and the one-way bearing.

4. The foldable electronic device according to claim **1**, wherein the fixing member has a hinge stop portion corresponding to the one-way bearing, the one-way bearing rotates relative to the rotating shaft in a second rotation direction opposite to the first rotation direction until the one-way bearing stops rotating after contacting the hinge stop portion.

5. The foldable electronic device according to claim **1**, wherein the one-way bearing comprises an inner ring member and an outer ring member, the inner ring member is disposed around the rotating shaft, the outer ring member is disposed around the inner ring member, and the outer ring member has the bearing stop portion.

6. The foldable electronic device according to claim **5**, wherein the outer ring member is tightly fitted to the inner ring member in the first rotation direction, and the outer ring member rotates relative to the inner ring member in a second rotation direction opposite to the first rotation direction.

7. The foldable electronic device according to claim **1**, wherein the first body rotates 360 degrees relative to the second body through the first hinge module.

8. The foldable electronic device according to claim **1**, wherein the first body rotates relative to the second body along the first rotation direction and unfolds to a first angle, such that the driving sheet is driven to rotate to the first position, and the back surface is located between the second body and the display surface.

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